Jaejun Park

Ph.D. Candidate | Human-Computer Interaction & Haptics
Department of Computer Science and Engineering, POSTECH
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Research Interests

- Multimodal Haptic Feedback (vibrotactile, thermal, and kinesthetic)
- Haptic Interaction in Virtual Reality
- LLMs and Automatic Haptic Generation
- Human-Computer Interaction

Education

2021.02 – Present **Ph.D. Candidate, Computer Science and Engineering**
Pohang University of Science and Technology (POSTECH)

2017.02 – 2021.02 **B.S., Computer Science and Engineering**
Pohang University of Science and Technology (POSTECH)

Publications

International Journal Papers

[J2] **Jaejun Park**, Soyeon Nam, Jeongwoo Kim, Uison Ju, and Seungmoon Choi, "GenTouchVR: Generating a Touchable Virtual Reality Environment from a Single Image," *IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026. (IEEE VR 2026 Track).*

[J1] Junwoo Kim, **Jaejun Park**, Chaeyong Park, Junseok Park, and Seungmoon Choi, "Representing Egocentric Directions with Torso-Applied Vibrotactile Stimuli," *IEEE Transactions on Haptics (TOH), 2026.*

International Conference Papers

[C7] Dajin Lee, **Jaejun Park**, Jeonggoo Kang, Taekun Yun, Dong-Chul Park, and Seungmoon Choi, "Participatory Design for In-Vehicle Vibrotactile Warnings on Driver's Seat," *EuroHaptics Conference (EH), 2024.*

[C6] Junwoo Kim, **Jaejun Park**, Chaeyong Park, Junseok Park, and Seungmoon Choi, "Human Identification Performance of Vibrotactile Stimuli Applied on the Torso along Azimuth or Elevation," *EuroHaptics Conference (EH), 2024.*

[C5] **Jaejun Park**, Sangyoon Han, and Seungmoon Choi, "Merging Camera and Object Haptic Motion Effects for Improved 4D Experiences," *IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2023.*

[C4] **Jaejun Park**, Junwoo Kim, Chaeyong Park, Sangyoon Han, Junseok Park, and Seungmoon Choi, "Information Transfer of Full-Body Vibrotactile Stimuli: An Initial Study with One to Three Sequential Vibrations," *IEEE World Haptics Conference (WHC), 2023.*

[C3] Sangyoon Han, **Jaejun Park**, and Seungmoon Choi, "Generating Haptic Motion Effects for Multiple Articulated Bodies for Improved 4D Experiences: A Camera Space Approach," *ACM CHI Conference on Human Factors in Computing Systems (CHI), 2023.*

[C2] **Jaejun Park**, Jeongwoo Kim, Chaeyong Park, Seungjae Oh, Junseok Park, and Seungmoon Choi, "A Preliminary Study on the Perceptual Independence Between Vibrotactile and Thermal Senses," *EuroHaptics Conference (EH), 2022.*

[C1] Jiwan Lee, **Jaejun Park**, and Seungmoon Choi, "Absolute and Differential Thresholds of Motion Effects in Cardinal Directions," *ACM Symposium on Virtual Reality Software and Technology (VRST), 2021.*

International Posters and Demonstrations

[D2] Jeongwoo Kim, **Jaejun Park**, Uison Ju, and Seungmoon Choi, "Automatic Haptic Rendering Pipeline Using AI Models for Generative Virtual Environments," *IEEE World Haptics Conference (WHC), 2025.*

[D1] Jiwan Lee, **Jaejun Park**, Jaehyeok Ahn, Sangyoon Han, Gyeore Yun, Jaebong Lee, Beomsu Lim, and Seungmoon Choi, "Automatic Generation of Haptic Motion Effects from Audiovisual Content," *IEEE World Haptics Conference (WHC), 2025.*

[W1] Junwoo Kim, **Jaejun Park**, Chaeyong Park, Junseok Park, and Seungmoon Choi, "Identification of Spatio-Temporal Vibrotactile Stimuli Across the Torso: Toward Egocentric Haptic Navigation," *IEEE World Haptics Conference (WHC), 2025. (Work-in-Progress).*

Domestic Conference Papers

[KC5] **Jaejun Park**, Soyeon Nam, Jeongwoo Kim, Uison Ju, and Seungmoon Choi, "GenTouchVR: An Automatic Generation Pipeline of Touch-Enabled Virtual Environments from a Single Image," *Korea Haptics Conference (KHC), 2026.*

[KC4] Junwoo Kim, **Jaejun Park**, Chaeyong Park, Junseok Park, and Seungmoon Choi, "Identification Performance of Vibration Locations Applied on the Torso along Azimuth or Elevation," *Korea Haptics Conference (KHC), 2024.*

[KC3] Junwoo Kim, **Jaejun Park**, Chaeyong Park, Junseok Park, and Seungmoon Choi, "Measurement of Perceived Elevation Angles of Vibrotactile Stimuli on the Torso," *Korea Haptics Conference (KHC), 2023.*

[KC2] **Jaejun Park**, Sangyoon Han, and Seungmoon Choi, "Merging Camera and Object Haptic Motion Effects for Improved 4D Experiences," *Korea Haptics Conference, 2023.*

[KC1] **Jaejun Park**, Junwoo Kim, Chaeyong Park, Sangyoon Han, Junseok Park, and Seungmoon Choi, "Estimating Information Transfer for Sequential Full Body Vibrotactile Stimuli," *HCI Korea, 2023.*

Domestic Posters and Demonstrations

[KD2] **Jaejun Park**, Juhyeop Lee, and Seungmoon Choi, "GenTouchVR Demo: Experiencing a Touch-Enabled Environment Automatically Generated from a Single Image," *Korea Haptics Conference, 2026.*

[KD1] Jiwan Lee, **Jaejun Park**, Jaehyeok Ahn, Sangyoon Han, Gyeore Yun, Jaebong Lee, Beomsu Lim, and Seungmoon Choi, "Automatic Authoring of Physical and Perceptual Motion Effects for Virtual Reality," *Korea Haptics Conference, 2024.*

Awards & Honors

2026	Best Demonstration Award – Honorable Mention [KD2]	KHC 2026
	Best Paper Award – Honorable Mention [KC5]	KHC 2026
2025	People's Choice Award [D2]	WHC 2025
	People's Choice Award [D1]	WHC 2025
2023	Best Paper Award – 2nd Place [KC2]	KHC 2023
	Best Paper Award [KC1]	KHCI 2023

Research Projects

24.06–Present **Interaction Framework for Connecting Users to Generative Experiences via Time-Space Expanded Multimedia**
National Research Foundation of Korea (NRF)

24.05-24.11 **Analysis of Information Transfer Capacity in Visual-to-Tactile Substitution**
Electronics and Telecommunications Research Institute (ETRI)

24.01–26.06 **Development of Tactile Standards and High-fidelity Integrated Haptic System for Hyper-realistic Metaverse**
National Research Council of Science & Technology (NST)

23.05-23.11 **The Perceptual Analysis of Multimodal Haptic Stimuli and Visual-to-Haptic Substitution**
Electronics and Telecommunications Research Institute (ETRI)

22.07-23.06 **Function Advancement to Improve the Marketability of Vibration Seats**
Hyundai Motor Company

22.05-22.11 **The Perceptual Analysis of Multimodal Haptic Stimuli and Visual-to-Haptic Substitution**
Electronics and Telecommunications Research Institute (ETRI)

21.05-21.11 **Analysis of tactile resolution for visual information delivery**
Electronics and Telecommunications Research Institute (ETRI)

21.02–23.06 **Automatic Authoring of Physical and Perceptual/Affective Motion Effects for Virtual Reality**
Samsung Research Funding & Incubation Center for Future Technology